

1. Overview

NOGAME is a mobile puzzle game where players progress through a sequence of levels. Player experience is strongly influenced by level difficulty: levels that are too easy may reduce engagement, while levels that are too difficult may create frustration and interrupt progression.

This analysis focuses on understanding level difficulty and player progression using gameplay event data, with the goal of identifying levels that may require balancing or additional player support.

2. Business Problem

The game team needs to identify levels that create excessive difficulty or progression friction so that balancing efforts can be prioritized effectively.

Rather than simply finding the hardest levels, the analysis needs to determine which levels are both difficult and meaningful to the player journey — for example, levels with low win rates, repeated attempts, high revive usage, or noticeable progression drop-off.

so , which levels should the game team prioritize for balancing, and what actions should be considered for those levels?

3. Analysis Objectives

The objective of this analysis is to identify levels that create excessive difficulty or progression friction and determine which levels should be prioritized for balancing.

The analysis aims to answer **three key questions**:

- Which levels show unusually high difficulty?
- Which difficult levels are associated with higher progression drop-off?
- Which levels should be prioritized based on difficulty, player exposure, and progression impact?

4. Data Scope

The analysis uses gameplay event data collected from `level_start` and `level_end`, which contain information about player progression, attempts, level outcomes, playtime, revive usage and other gameplay attributes.

The analysis primarily focuses on the **first 500 progression levels**, where `level_id` and `level_index` are highly consistent and the player sample size is more reliable.

5. Data Quality & Assumptions

Before analysis, several checks were performed to validate the event data and clarify field interpretation.

status = 1 was treated as a successful level completion, while other status values were treated as non-win outcomes.

The dataset contains repeated level content at later progression positions, therefore `level_id` and `level_index` were analyzed separately.

Some duplicated event keys were observed, so the analysis mainly relies on aggregated level-level metrics rather than assuming a perfect one-to-one event match.

6. Methodology

Level difficulty was evaluated using multiple gameplay metrics rather than a single indicator.

The **main metrics** include:

- First-attempt Win Rate
- Attempt Win Rate
- Average Attempts
- Average Playtime
- Revive Rate
- Progression Drop-off Rate
- Users Reached

A level is considered more concerning when it is difficult, affects a meaningful number of players, and is associated with a noticeable progression drop-off.

7. Key Findings

Finding 1: Difficulty spikes exist at specific levels

Several levels show clear difficulty spikes compared with the general level population.

Levels 309, 304, 434 and 394 have particularly low first-attempt win rates, suggesting that players frequently fail on their first exposure to these levels.

Finding 2: High difficulty does not always mean players immediately abandon

Retry-after-failure rates remain very high for the shortlisted difficult levels.

For example, Level 309 has the lowest first-attempt win rate among the priority levels, but players still retry at a very high rate.

This suggests that the level creates friction, but the difficulty does not necessarily translate directly into immediate abandonment.

Finding 3: Some levels combine difficulty with progression friction

Levels such as 304 and 434 are more concerning because they combine low win rates with relatively high progression drop-off.

These levels may create a stronger interruption in the player journey and therefore deserve higher balancing priority.

8. Priority Levels

Level	First-attempt Win Rate	Avg Attempts	Avg Playtime	Revive Rate	Key Concern
231	31.5%	6.78	111.7s	36.4%	High retry burden, broad exposure
304	21.1%	5.42	67.6s	42.0%	Strong difficulty spike
309	15.7%	5.29	119.9s	54.2%	Lowest first-try

					success, high revive
434	22.8%	5.20	143.8s	52.8%	Difficult and time-consuming

These four levels were selected because they show the strongest combination of difficulty, player effort and progression impact.

9. Recommendations

Level 231

Test a moderate difficulty reduction or provide additional player support. The level affects a relatively large player base and has the highest average number of attempts among the priority levels.

Level 304

Test a small increase in move limit or simplify part of the level layout. The goal should be to reduce excessive friction without removing the intended challenge.

Level 309

Avoid a large nerf at first. Players continue retrying despite the high difficulty, so a better first test may be improved hint support, revive assistance or a small balancing adjustment.

Level 434

Reduce complexity or shorten the level experience. The combination of low win rate, high revive usage and long playtime suggests high player effort.

Recommended changes should be validated through controlled experiments. Key evaluation metrics should include completion rate, attempts per completion, progression to the next level and downstream retention.

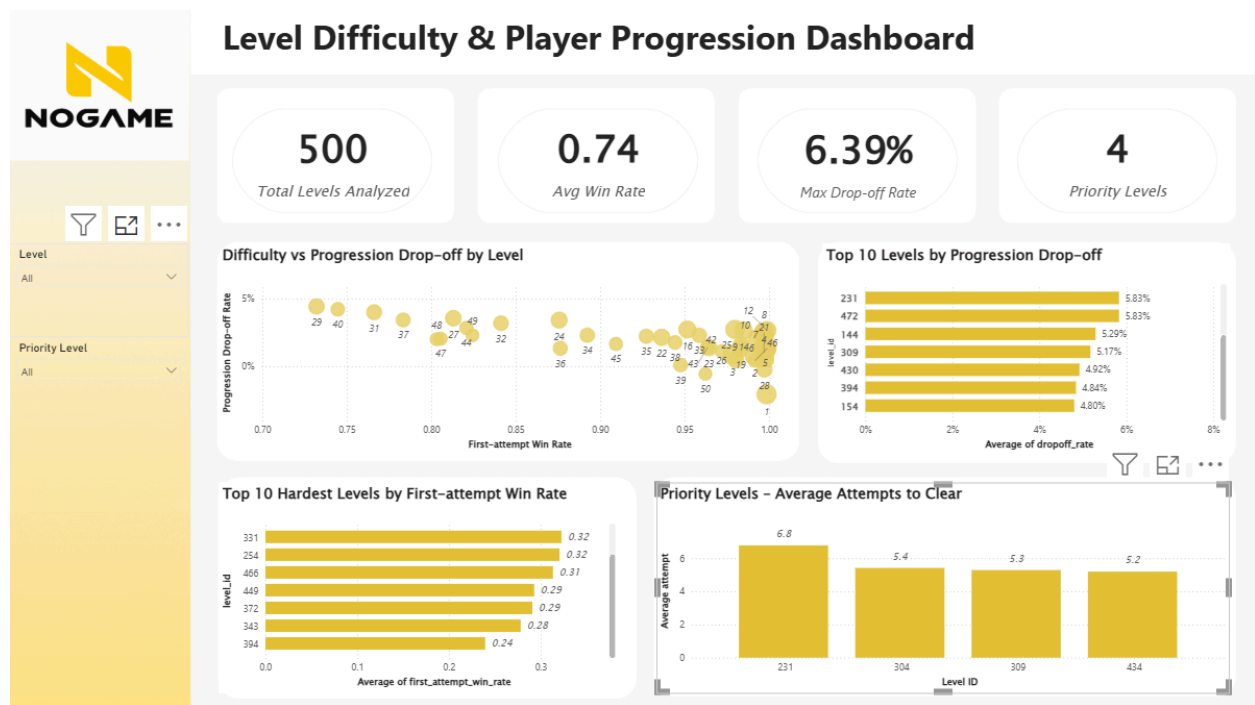
10. Limitations

This analysis identifies associations between difficulty and player progression, but it does not prove that a specific level directly causes churn.

The dataset also covers a limited observation window, so some players may enter the dataset after having already completed earlier levels.

This may create minor inconsistencies in progression-based metrics, especially in early levels.

11. Dashboard



Level Difficulty & Player Progression Dashboard

The dashboard allows the game team to interactively explore level difficulty, progression drop-off and priority levels.